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Reg. No. | | | | | | | | | | |

[13CE53]

FIFTH SEMESTER B.Tech/B.E EXAMINATION, DECEMEBER-2015

FOUNDATION ENGINEERING

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full questions.

1. a. Discuss with neat sketches any two boring methods used in soil exploration. **08**

   b. The inner diameters of a sampling tube and that of the a cutting edge are 70mm and 68mm respectively, their outside diameters are 72 and 74mm respectively. Determine the **06**

   i. Inside clearance

   ii. Outside clearance

   iii. Area ratio of the sampler.

2. a. Write an explanatory note on the general types foundations, with suitable sketches. **08**

   b. What are the conditions under which combined footings are used? **06**

3. a. Explain how 'water table' affects the Bearing capacity of the soil. **06**

   b. A square footing located at a depth of 1.3m below the ground has to carry a safe load of 800kN. Find the size of the footing if the desired FOS=3. The soil has the following properties: **08**

   Voids ratio=0.55, degree of saturation =50% specific gravity =2067,

   C=8KN|m^2 $\phi = 30^\circ$

   $N_{C=}37.2,\ N_q = 22.5\, and\ N_\gamma = 19.7$

   Use Terzaghi's analysis.

4. a. How do you find load carrying capacity of the piles. **06**

   b. How are piles classified based on various factors? Write at least two characteristics of each class. **08**

5. a. Explain the different modes of vibration of a foundation block subject to dynamic forces from a machine. **06**

   b. Describe with neat sketch "Mass Spring System". How do you calculate natural frequency of foundation soil system. **08**

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6. a. Why do foundations settle? Explain. **04**

b. A rectangular footing 2mx3m carries a column load of 600KN at a depth of 1m. The footing rests on a C-ϕ soil strata 6m thick, having Poisson's ratio of 0.25 and E=20000 KN|m^2. Calculate the immediate settlement of a footing. (Take, I_w=1.06) **05**

c. A soft, normally consolidated clay layer is 6m thick with natural water content of 30%. The clay has $\gamma_{sat} = 17.4\,KN|m^3, G = 2.67$ and liquid limit of 40%. The ground water table is at the surface of the clay. Determine the settlement of the foundation if the foundation load will subject the center of the clay layer to vertical stress increase of 8KN|m^2. **05**

7. a. Write the difference between active and passive earth pressure. **04**

b. A retaining wall 4m high, has a smooth vertical back. The backfill has a horizontal surface in level with the top of the wall. There is uniformly distributed surcharge load of 36 KN|m^2 ; its angle of shearing resistance is 30^0 and cohesion is zero Determine the magnitude and point of application of active earth pressure per meter length of the wall. **10**

8. a. Write a short note on uses of coffer dams. **04**

b. Explain briefly the procedure describing the design of cantilever anchored sheet pile walls. (Fixed earth method). **04**

c. Rankine's Theory **06**